# NYAYMITRA 2.0 — Prototype Progress and Evidence Report

Implemented, tested, demonstrated, incomplete, test outputs, commit references

## NYAYMITRA 2.0 — Prototype Progress and Evidence Report

### 1. What has been implemented

- Backend: FastAPI app factory, config, security, middleware, logging (PII redaction), models, routers, services
- AI/RAG: provider abstraction (openai_compatible + offline), retrieval, citation validation, safety guard, registry
- Frontend: citizen SPA + admin console (English/Gujarati, accessible)
- Database: SQLite, 4 migrations, seed data (9 articles, 11 departments, 5 triage guides, 18 glossary terms), 2,125 quarantined legacy rows
- Scripts: init_db, check_db, backup_db, purge_data, ingest_constitution, constitution_coverage, license_audit, verify_ai_provider, verify_changeset, smoke_web
- Deployment: Dockerfile, systemd unit, render.yaml, .env.example
- Documentation: security, privacy, IP, legal, setup, audit, architecture, API, schema, testing

### 2. What has been tested

| | |
|---|---|
| `pytest` full suite | 226 passed, 0 failed |
| Coverage `--cov=backend/app` | 92% (2,545 statements, 202 missed) |
| `pip-audit` dependencies | No known vulnerabilities |
| `scripts/check_db.py` | OK (integrity ok, 4 migrations, 0 missing indexes) |
| `node --check` frontend | Pass |
| `scripts/verify_ai_provider.py` | Exit 3 — no model runtime (BLOCKED) |
| jsdom smoke test (live server) | All citizen journeys + admin console OK |
| Frontend static checks | Pass |

### 3. What has been demonstrated

Against a live `uvicorn` server on `0.0.0.0:8000`:

- Home, assistant (answer + citations + disclaimer), library search, glossary filter
- Language toggle EN ↔ GU persists
- Complaint: preview → consent checkbox → submission → reference + access token (NYM-XXXXXX)
- Saved-complaint lookup by reference + token; wrong token → generic 404 (no oracle)
- Admin console: stats, verification queue, metrics, documents
- All routes return expected status; CSP/security headers present

### 4. What remains incomplete

| | | |
|---|---|---|
| Official Constitution source documents | Not loaded | B-1: owner must supply PDFs |
| Human legal review of articles/departments/guides | 0 of 9 verified | B-2: reviewer not appointed |
| Real AI model inference | Not available | B-4: no Ollama/model runtime |
| Real-device / screen-reader testing | Not done | B-3: no browser/device in sandbox |
| Public deployment | Not deployed | B-5: owner approval required |

## 5. Test outputs

``` 226 passed, 1 warning in ~20s TOTAL 2545 statements 202 missed 92% pip-audit → No known vulnerabilities found PRAGMA integrity_check = ok; 4 migrations applied, 0 missing indexes /scripts/verify_ai_provider.py → exit 3 (no provider configured) ```

## 6. GitHub commit references

- Repository: https://github.com/dhundhwana-arch/Nyaymitra-2.0
- Branch: `arena/812d6894-nyaymitra-2-0`
- Current repository commit: `cc56d0e`
- Commit used to generate PDFs: `cc56d0e` (rebuilt after remediation)
- Commit on which tests were executed: `792b529` (226 passed)
- Commit associated with CI evidence: `cc56d0e` (GitHub push run `42595255121`, success); prior push `42594844444` on `792b529` also succeeded
- Prior commit: `792b529` (source validation commit; superseded by `cc56d0e`)
- Earlier: `5a6dcb9` (AI reliability, usage tracking, review packet, deployment configs)
- PR #1: https://github.com/dhundhwana-arch/Nyaymitra-2.0/pull/1 (open, draft, mergeable; head = `cc56d0e`)

## 7. Screenshots

No screenshots are included. Behaviour was verified against a live server via the `scripts/smoke_web.mjs` jsdom test; all journey output is recorded above. No fabricated or invented screenshots are used.

## 8. Repository

https://github.com/dhundhwana-arch/Nyaymitra-2.0

NYAYMITRA 2.0 — functional software prototype / Proof of Concept — subject to independent evaluation. Not legal advice; not a replacement for courts, lawyers, police, or government authorities. https://github.com/dhundhwana-arch/Nyaymitra-2.0 (branch arena/812d6894-nyaymitra-2-0, commit cc56d0e)
```